Department of Computer Science and Engineering
Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch	**:** III Semester B.E CSE B
Course Code & Title	**:** CS3352 & Foundations of Data Science
Name of the Faculty member	**:** Mrs. P.Sabeena Burvin, AP / CSE

Innovative Practice Description

- **Unit / Topic: Unit I / Facets of Data**
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO 1,2,10,11
- **Activity Chosen:** Flipped Classroom
- **Date of Implementation:** 01.08.2025
- **Justification:**
  The flipped classroom approach involves students learning fundamental concepts independently before class through videos, readings, or tutorials and then using class time for discussion, problem-solving, and hands-on activities. For this session, the focus was on understanding the **different facets of data** that form the core of data science. The pre-class and in-class activities aimed to help students grasp the nature, quality, structure, and lifecycle of data.
- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**
  Pre-Class Learning
  Resources Provided:
  - Video lecture: https://www.youtube.com/watch?v=vVZdwBFRF7Y
  - Reading material: https://canvas.instructure.com/courses/12583558

Key Concepts Learned:

1. **Types of Data:**
   - Structured Data**:** Organized in rows and columns (e.g., databases).
   - Unstructured Data**:** Includes text, images, audio, video, etc.
   - Semi-Structured Data: partially organized but flexible.
2. **Data Quality Dimensions:**
   - Accuracy, completeness, consistency, timeliness, validity, and uniqueness.
3. **Data Lifecycle:**
   - Collection → Storage → Processing → Analysis → Visualization → Archival.
4. **Sources of Data:**
   - Machine-generated (IoT, sensors), human-generated (social media), and transactional data.
5. **Big Data Characteristics (5Vs):**
   - Volume, Velocity, Variety, Veracity, and Value

In-Class Activities:

Activity: *Data Classification Exercise*

- Students were divided into groups and given mixed datasets (e.g., tweets, sales records, sensor logs).
- Each group classified the data as structured, semi-structured, or unstructured and discussed suitable processing techniques.
- One student from each group explained various structured types for their allotted data sets.

Learning Outcomes:

By the end of the session, students were able to:

- Identify and describe the different facets and types of data in data science.
- Evaluate data quality and understand its impact on analysis outcomes.
- Apply the concept of data lifecycle to real-world data management.
- Discuss ethical implications and governance of data usage.

- **CO – PO / PSO mapping:**

CO	**PO1**	**PO2**	**PO4**	**PO9**	**PO10**	**PO12**	**PSO1**	**PSO3**
CO1	**2**	**2**	**1**	**3**	**3**	**2**	**3**	**2**

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO4**	**PO9**	**PO10**	**PO12**
	2	**2**	**1**	**3**	**3**	**2**
Justification for correlation	Apply basic Knowledge and identify the data facets	Identify the classification of data in Data Science process	Able to identify the structure of various kinds of data	Functional individually in identifying the representation of data	Communicate / share the ideas with other students	Ability to classify the data contents gathered through self-learning

- **Images / Screenshot of the practice:**

- **Reflective Critique:**

  - ***Feedback of practice from students and other stakeholders:***
    - Student felt good, since they can study at their own pace/time.
    - They felt that through such learning, they can explore more.
  - ***Benefit of the practice:***
    - More collaboration time for students.
    - Students learn at their own pace.
    - Rather than learning in a traditional classroom setting, the flipped classroom uses a more application-based approach for students.
    - It encourages students to come to class prepared.
  - ***Challenges faced in implementation:***
    - Relies on student preparation – few students did not come prepared.
    - The depth of the subject can be dictated by the student themselves or the group the student is working with.
    - The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- https://bokcenter.harvard.edu/flipped-classrooms
- https://en.wikipedia.org/wiki/Flipped_classroom
- https://www.youtube.com/watch?v=wwlgsg2Igc